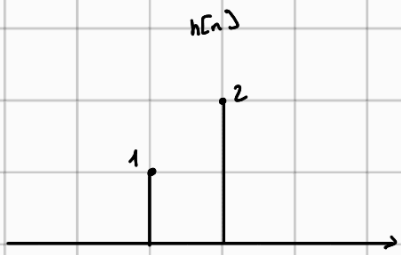
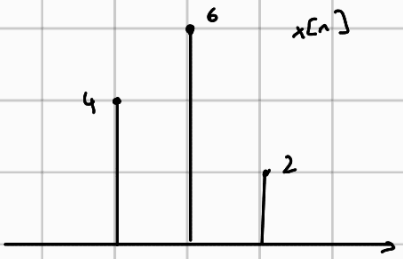
Quiz çözümü

$x(t) = \cos(0{,}1\pi t + 0{,}2)$

$f_s = 5\,Hz \quad x[2] = ?$

$x[2] = \cos(0{,}1\pi \cdot \frac{2}{5} + 0{,}2)$



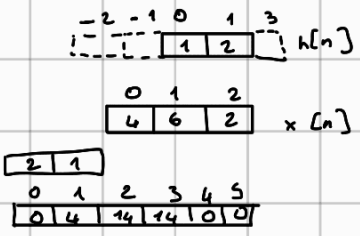


$H(z) = 1 + 2z^{-1}$

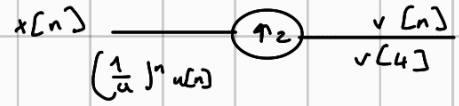
$X(z) = 4 + 6z^{-1} + 2z^{-2}$

$X(z)\ H(z) \Rightarrow$

Başka bir yol



3. soru



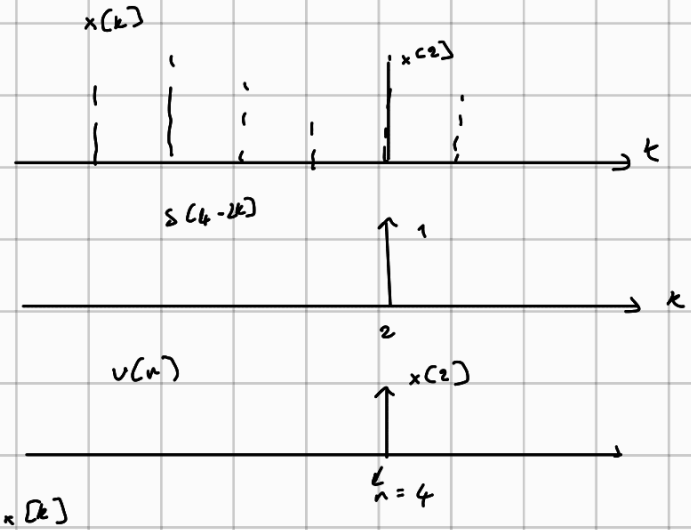
x[n]

1	2	3	4	5	

v[n]

1	0	2	0	3	0

$\left(\frac{1}{a}\right)^2$

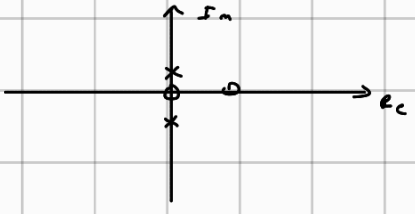


$v[n] = \sum_{k=-\infty}^{\infty} \delta[n-kL]\, x[k]$

$v[4] = \delta[4-2\cdot 2]\, x[2] = \sum_{k=-\infty}^{\infty} \delta[4-2k]\, x[k]$

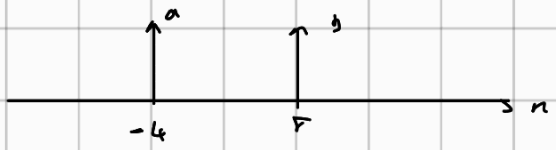
$X(z) = \frac{1-0{,}81z^{-1}}{1+0{,}25z^{-2}} \quad |z| > 0{,}5$

$= \frac{z^2 - 0{,}81z}{z^2 + 0{,}25} = \frac{z(z-0{,}81)}{(z+j0{,}5)(z-j0{,}5)}$



$$h[n] = \cos\left(\frac{\pi}{2} n\right)\left[\delta[n+4] - \delta[n-8]\right]$$

$$h[n] = \underbrace{\cos\left(\frac{\pi}{2}(-4)\right)}_{a}\,\delta[n+4] \;\underbrace{-\cos\left(\frac{\pi}{2} 8\right)}_{b}\,\delta[n-8]$$



$$H(z) = a z^{4} + b z^{-8} = a z^{4} + \frac{b}{z^{8}}$$

$$X(z) = \frac{0,4}{1 - 3 z^{-1}} \qquad |z| < 3$$

$$x[n] * \delta[n+2] = ?$$

$$(0,4)(-3^{n}\, u[-n-1])$$

$$X(z) = \frac{1}{1 - a z^{-1}} \qquad |z| < a$$

$$x[n] = -a^{n}\, u[-n-1]$$

IIR Filter Design by the B.T.

$$H(s) = \frac{b}{s+a} = \frac{Y(s)}{X(s)} \qquad H(s) = \frac{1}{s^3 + 2s^2 + 3s + 4} = \frac{k_1}{s-a} + \frac{k_2}{s-b} + \cdots$$

$$\frac{dy(t)}{dt} + a\, y(t) = b\, x(t) \quad (*)$$

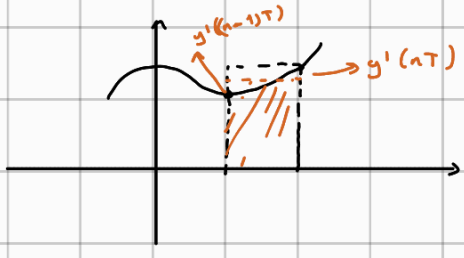
$$y'(t) + a\, y(t) = b\, x(t)$$

$$y'(t) = b\, x(t) - a\, y(t)$$

$t \leftarrow nT$ $\qquad y'(nT) = b\, x(nT) - a\, y(nT)$ $\qquad (1)$

$$y(t) = \int_{t_0}^{t} y'(t)\, dt + y(t_0)$$

$t \leftarrow nT$, $t_0 \leftarrow (n-1)T$ $\qquad y(t) = \int_{t_0}^{t} y'(t)\, dt + y(t_0)$



$$y'(nT) = \frac{T}{2}\left[y'(nT) + y'((n-1)T)\right] + y((n-1)T) \quad (2)$$

Substitute (1) into (2)

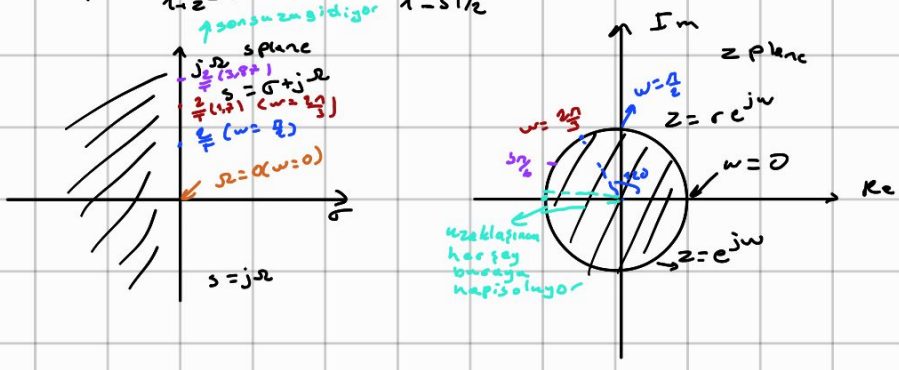
$$(1 + a\tfrac{T}{2})\, y[n] + (-1 + a\tfrac{T}{2})\, y[n-1] = b\tfrac{T}{2}\left[x[n] + x[n-1]\right]$$

(

$$H(z) = \frac{Y(z)}{X(z)} = b \frac{1}{2} \frac{1+z^{-1}}{\left(1+a\frac{T}{2}\right)+\left(-1+a\frac{T}{2}\right)z^{-1}} = \frac{b}{\frac{2}{T}\left(\frac{1-z^{-1}}{1+z^{-1}}\right)+a} \qquad H(s) = \frac{b}{s+a}$$

$$H(z) = H(s)\Big|_{s \leftarrow \frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}}$$

$$s = \frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}} \qquad z = \frac{1+sT/2}{1-sT/2}$$



s yerine $j\Omega$ z yerine $e^{j\omega}$

$$j\Omega = \frac{2}{T}\frac{e^{j\omega}-1}{e^{j\omega}+1} = \frac{2}{T}\frac{e^{j\frac{\omega}{2}}\left(e^{j\frac{\omega}{2}} - e^{-j\frac{\omega}{2}}\right)}{e^{j\omega/2}\left(e^{j\omega/2} + e^{-j\omega/2}\right)} = \frac{2}{T}\frac{2j\sin\left(\frac{\omega}{2}\right)}{2\cos\left(\frac{\omega}{2}\right)} = j\frac{2}{T}\tan\left(\frac{\omega}{2}\right)$$

$$j\Omega = j\frac{2}{T}\tan\left(\frac{\omega}{2}\right)$$

$$\Omega = \frac{2}{T}\tan\left(\frac{\omega}{2}\right)$$

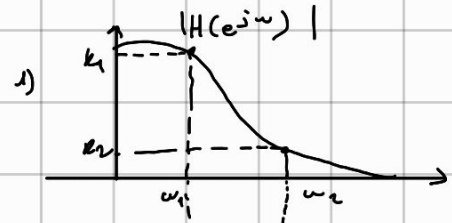
~~Ω ... ω~~

$$\omega = 2\arctan\left(\frac{\Omega \cdot T}{2}\right)$$

Örnek

Specleri belirleyelim

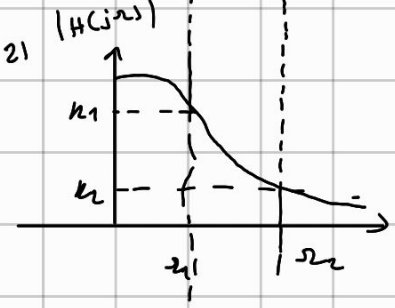
1)



$k_1 @ \omega_1$

$k_2 @ \omega_2$

2)



$k_1 @ \Omega_1 = \frac{2}{T}\tan\left(\frac{\omega_1}{2}\right)$

$k_2 @ \Omega_2 = \frac{2}{T}\tan\left(\frac{\omega_2}{2}\right)$

3) design $H(s)$

4) $H(z) = H(s)\Big|_{s \leftarrow \frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}}$

EX: Design LPF, B.T → sadece Butterworth



a) Monotonic stop band, pass band

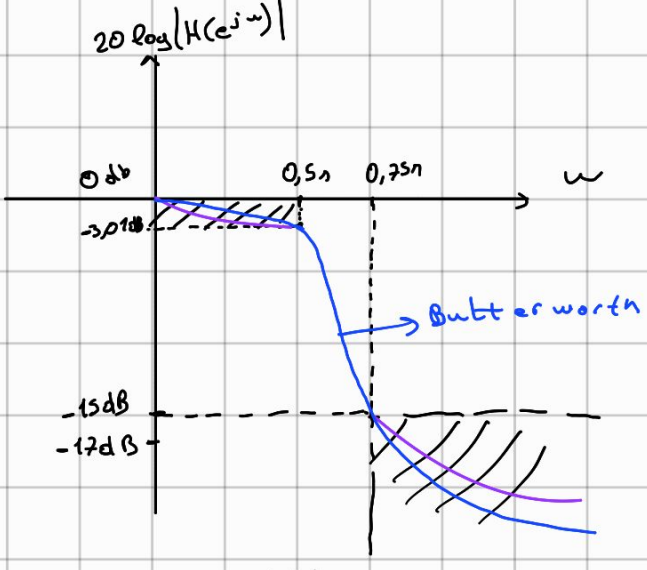
b) -3,01 dB cut off at 0,5π rad

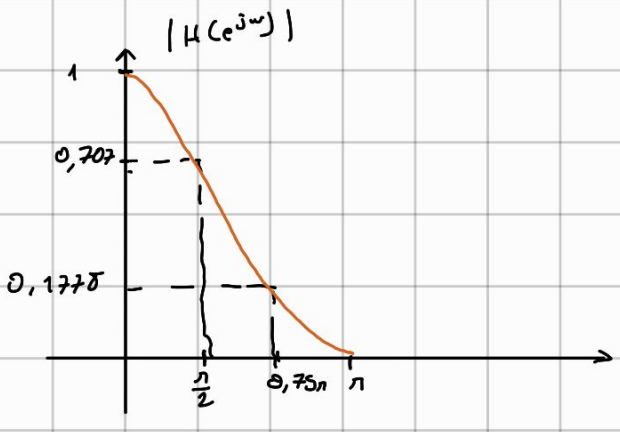
c) 0,75π rad ⇒ at least 15 db olsun

$F_s = 100\,Hz$

$T = 10^{-3}\,sec$

$\frac{1}{T} = F_s$





$20 \log |H(e^{j\omega})| = -3,01\,dB$

$20 \log |H(e^{j\omega})| < -15\,dB$

$F_s = 1000\,Hz$

$T = 10^{-3}\,sec$

$\Omega_1 = \frac{2}{T} \tan\left(\frac{\omega_1}{2}\right)\Big|_{\omega_1 = 0,5\pi}$ k_1 @ ω_1

$\Omega_2 = \frac{2}{T} \tan\left(\frac{\omega_2}{2}\right)\Big|_{\omega_2 = 0,75\pi}$

k_1 @ $\Omega_1 = \frac{2}{T} \tan\left(\frac{\omega_1}{2}\right)$

k_2 @ $\Omega_2 = \frac{2}{T} \tan\left(\frac{\omega_2}{2}\right)$

$\Omega_1 = 2000$ rad/sec

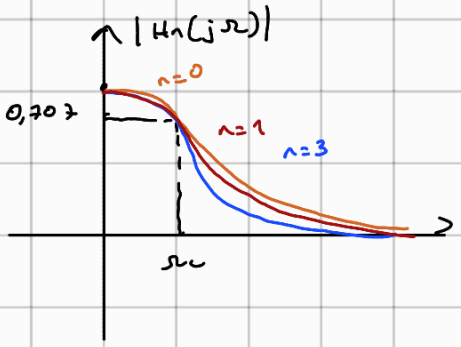
$\Omega_2 = 4828,42$ rad/sec

$k_1 = 0,707$ $k_2 = 0,1778$

Butterworth

$$|H_n(j\Omega)|^2 = \frac{1}{1+\left(\frac{\Omega}{\Omega_c}\right)^{2n}}$$

n neden bağımsız olarak Ω_c için geçilen $\frac{1}{\sqrt{2}}$



$$|H_n(j\Omega)|^2 = \frac{1}{1+\left(\frac{\Omega}{\Omega_c}\right)^{2n}}$$

$$|H_n(j\Omega_n)|^2 = \frac{1}{1+\left(\frac{\Omega_n}{\Omega_c}\right)^{2n}}$$

① $|H_n(j\Omega_1)|^2 = \frac{1}{1+\left(\frac{2000}{\Omega_c}\right)^{2n}} = (0,707)^2$ $\Rightarrow \Omega_c = 2000$ rad/sec

② $|H_n(j\Omega_2)|^2 = \frac{1}{1+\left(\frac{\Omega_2}{\Omega_c}\right)^{2n}}$

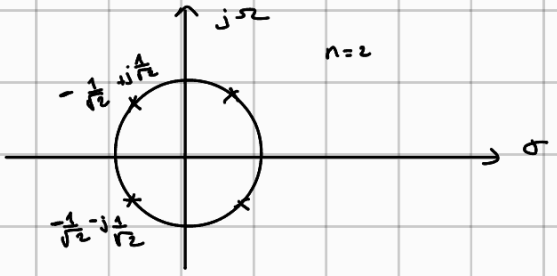
$$(0,1778)^2 \geq \frac{1}{1+\left(\frac{4828,42}{2000}\right)^{2n}}$$

$\Rightarrow n \geq 1,942$

n: 2, 3, 4 ... seçebilirim

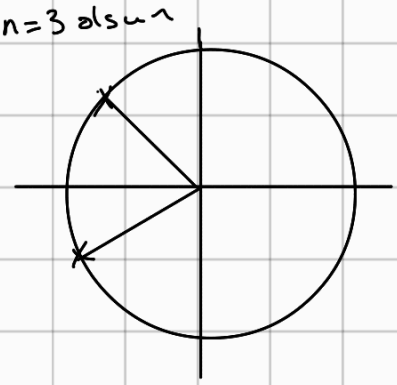
n = 2 seçelim

$$H(s) = \frac{k}{(s-s_0)(s-s_1)\cdots(s-s_{n-1})} = \frac{k}{\left(s-\left(-\frac{1}{\sqrt{2}}+j\frac{1}{\sqrt{2}}\right)\right)\left(s-\left(-\frac{1}{\sqrt{2}}-j\frac{1}{\sqrt{2}}\right)\right)}$$



$$= \frac{\Omega_c^2}{s^2+\sqrt{2}\,\Omega_c s+\Omega_c^2}$$

n=3 adsun



Tabdo

n	
1	$s+1$
2	$1+\sqrt{2}\,s+s^2$
3	$(s^2+s+1)(s+1)$

$n=2$

$$H(s)=\frac{1}{1+\sqrt{2}\,s+s^2}$$

$$H(s)=\frac{k}{s^2+\Omega_c\sqrt{2}\,s+\Omega_c^2}$$

$\Omega_c = 2000$

$$H(z)=H(s)\Big|_{s\in 2000\ \frac{1-z^{-1}}{1+z^{-1}}}$$

$$H(z)=\frac{1+2z^{-1}+z^{-2}}{3{,}4142+0{,}58\,z^{-2}}$$